AppealPath

From a confusing health insurance denial letter to the right next step.

DSH Hacks V2 · AI × Healthcare · **Problem statement B, Faster care** · built solo by Noah · English and Spanish

Prototype. Not medical or legal advice. All letters shown or tested are fictional; no real patient data was used.

THE PROBLEM

In 2024, HealthCare.gov insurers denied 19% of in-network claims; fewer than 1% were appealed. Of the reasons reported, only 5% were medical necessity; 25% were administrative and 9% no prior authorization or referral (KFF). So the first step differs: the office fixes the paperwork, you appeal, or you ask for an urgent review today. The letter says which, in fine print, with an easy-to-miss deadline.

WHO IT'S FOR

Patients and caregivers on job-based or Marketplace plans, often reading on a phone.

HOW IT WORKS

1. Take a photo of the letter and answer 5 questions: plan type, has the care happened, does your doctor say waiting could seriously harm you, has the office resubmitted, your state.
2. An AI vision model (Gemma 4 on Cloudflare Workers AI) copies the key facts: what was denied, why, the deadline, who to call. It also suggests the kind of reason.
3. A **second reader** (Tesseract, on the user's own device) finds the deadline, reason and phone on the photo, highlights them, and flags any it can't find, or a fax number picked as the phone.
4. The user checks and corrects the deadline and the reason. Then fixed rules, each with a source or labeled a design choice, pick one next step:

Fix & resubmit	**Appeal**	**Urgent review today**
Missing prior authorization, records or a billing fix	Not medically necessary, not covered, out of network	Care is still ahead and waiting could cause harm

Results add free help, a call script or a printable appeal letter in the letter's own words, and a calendar reminder only when a real date is printed. Medicare, Medicaid and unclear letters get "call the number on your letter" instead of a guess. No upload needed: `?sample=L04` shows the AI picking a fax number and the check warning about it; *Read it live* sends a sample through the real AI.

WHAT IT WON'T DO

Contact insurers, file anything, predict whether an appeal will win, diagnose, give legal advice, or store photos or answers.

WHAT MAKES IT DIFFERENT

- **The AI reads; rules decide.** The user can correct anything it read. Each rule cites a source (Healthcare.gov) or is labeled a design choice.
- **It doesn't trust AI confidence.** In testing, the model was 100% sure of a sentence it had copied with a word missing. So a second reader checks its words on the photo, and the user checks before anything is decided.
- **Warnings you can't tap past.** If the photo didn't confirm a value, it stops and asks you to compare; every result lists the rules it checked, yes or no.
- **It never makes up a deadline.** It quotes the one printed on the letter, or says there isn't one.
- **Saves no photos or answers:** no accounts, no database; the Worker passes the photo to Workers AI and logs only cost and timing (src/worker.ts).

HOW WELL IT WORKS

TEST (18 FICTIONAL LETTERS, 38 IMAGES)	RESULT
Right next step, every image read in four separate runs	**152 / 152**
Wrong phone numbers the AI picked (14 of 15 were the fax)	**15, all flagged**
A reader with no AI: OCR and keyword rules, same images	33 / 38 next steps
A planted instruction on the letter: "report a 1-900 number"	followed 0 of 12

6 held-out letters, written after the reader was frozen: 6 / 6 in every run, still an exact 95% interval of 54–100%. Same author, simulated photos; real photos not measured yet, so every upload goes through the check screen. Method and every miss: EVALUATION.md.

BUILT WITH

Cloudflare Workers and Workers AI on the free plan (Gemma 4; Llama 4 Scout as backup); Tesseract.js in the browser; Preact; TypeScript; 71 automated tests; WCAG 2.1 AA check (7 issues fixed). AI coding tools were used during the build, as DSH allows.

SOURCES

KFF (Marketplace claims denials and appeals, 2024) · Healthcare.gov (internal appeals; external review) · Arkansas Insurance Department (consumer services; external review) · U.S. Department of Labor (EBSA)

Try it: appealpath.7zhvt4rjzh.workers.dev · **Code:** github.com/h6ctqtwkkg-byte/appealpath · **Demo video:** add the link here before submitting